

# Package ‘breedRBayes’

September 3, 2026

**Type** Package

**Title** ASReml-Style Formula Front-End for the BGLR Bayesian Engine

**Version** 0.0.0.9000

**Description** A formula-based interface, inspired by 'asreml' and 'sommer', for fitting Bayesian mixed models with the 'BGLR' engine. Supports genomic prediction (GBLUP), random-regression / reaction-norm models, multi-trait models and factor-analytic structures. Provides extraction of posterior chains, variance components, heritabilities as full posterior distributions, and Markov-chain convergence diagnostics.

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**Encoding** UTF-8

**LazyData** false

**Depends** R (>= 4.1.0)

**Imports** BGLR,  
coda,  
RhpcBLASctl,  
stats,  
utils,  
ggplot2

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EnvRtype,  
parallel,  
testthat (>= 3.0.0),  
knitr,  
rmarkdown

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|         |                                                                  |
|---------|------------------------------------------------------------------|
| as_mcmc | <i>Convert a fitted model's posterior draws to a coda object</i> |
|---------|------------------------------------------------------------------|

---

**Description**

Reads the variance-component chains (and optionally the intercept mu) written by BGLR and returns them as a [coda::mcmc.list](#) (one component per chain), ready for [mcmc\\_diag\(\)](#) or coda/bayesplot functions.

**Usage**

```
as_mcmc(x, ...)

## S3 method for class 'breedRB_fit'
as_mcmc(x, what = "varcomp", ...)
```

**Arguments**

|      |                                                        |
|------|--------------------------------------------------------|
| x    | A breedRB_fit.                                         |
| ...  | Unused.                                                |
| what | Which quantities to return: any of "varcomp" and "mu". |

**Value**

A [coda::mcmc.list](#).

---

|       |                                                                                    |
|-------|------------------------------------------------------------------------------------|
| bbglr | <i>Fit a Bayesian mixed model with the BGLR engine using asreml-style formulas</i> |
|-------|------------------------------------------------------------------------------------|

---

## Description

Fit a Bayesian mixed model with the BGLR engine using asreml-style formulas

## Usage

```
bbglr(
  fixed,
  random = NULL,
  residual = NULL,
  data,
  relmat = list(),
  nIter = 5000,
  burnIn = 1000,
  thin = 5,
  nChains = 1,
  seed = 123,
  saveAt = NULL,
  verbose = TRUE
)
```

## Arguments

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fixed               | Two-sided formula for the fixed/mean part, e.g. <code>yield ~ 1 + env</code> . Use <code>cbind(t1, t2) ~ ...</code> for a multi-trait model.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| random              | One-sided formula of random terms, e.g. <code>~ vm(gen, G)</code> . Special functions: <code>vm(f, K)</code> genomic effect with relationship matrix <code>K</code> ; <code>mrk(f, M, method)</code> genomic effect from a <b>marker matrix</b> <code>M</code> that automatically fits GBLUP or RR-BLUP, whichever is cheaper (method is "auto", "GBLUP" or "RRBLUP"; "auto" picks GBLUP when markers $\geq$ genotypes, else RR-BLUP — the two are prediction-equivalent). Use <code>solve_SNP()</code> to obtain marker effects. <code>leg(x, n) / rr(x, n)</code> random regression of order <code>n</code> ; <code>fa(f, k) / rrc(f, k)</code> factor-analytic. Bare names are factors; <code>a:b</code> is an interaction. |
| residual            | One-sided residual formula. <code>NULL/~units</code> = homogeneous; <code>~dsum(~units   env)</code> = heterogeneous residual variances by env.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| data                | A data frame.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| relmat              | Named list of matrices referenced by <code>vm()</code> and <code>mrk()</code> . For <code>vm()</code> these are relationship / covariance matrices ( <code>K</code> , not its inverse), with row/column names matching the factor levels. For <code>mrk()</code> the entry is a marker matrix with genotypes in rows (row names = genotype IDs) and markers in columns.                                                                                                                                                                                                                                                                                                                                                        |
| nIter, burnIn, thin | MCMC controls passed to BGLR.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|         |                                                                                                      |
|---------|------------------------------------------------------------------------------------------------------|
| nChains | Integer number of independent chains (distinct seeds). Enables Gelman-Rubin diagnostics when $> 1$ . |
| seed    | Base random seed; chain $c$ uses $\text{seed} + c - 1$ .                                             |
| saveAt  | Directory for BGLR binary output. Defaults to a fresh temp dir.                                      |
| verbose | Logical; print BGLR progress.                                                                        |

### Value

An object of class `breedRB_fit`.

### Examples

```
G <- readRDS(system.file("extdata", "kinship.matrix.rds", package = "breedRBayes"))
dat <- read.csv(system.file("extdata", "data_soy.csv", package = "breedRBayes"))
fit <- bbglr(yield ~ 1 + env, random = ~ vm(gen, G), data = dat,
             relmat = list(G = G), nIter = 2000, burnIn = 500, nChains = 2)
```

---

gebv

*Posterior genomic estimated breeding values (GEBV)*

---

### Description

**Deprecated.** Use `solution()`, which extracts posterior solutions for any term (genomic `vm()` BLUPs, plain random-factor BLUPs, and fixed effects). `gebv(fit, term)` is equivalent to `solution(fit, term, type = "random")` with the value column renamed `gebv`.

It still works for a `vm()` genomic term for now, mapping the ridge coefficients back through the PC rotation (genetic value =  $\text{PC} \% \% \text{b}$ ).

### Usage

```
gebv(fit, term = NULL, prob = 0.95)
```

### Arguments

|      |                                                                                                                                                            |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fit  | A <code>breedRB_fit</code> (single-trait).                                                                                                                 |
| term | Genetic term identifier (label or key). Defaults to the first <code>vm()</code> term. For a non-genomic random effect use <code>solution()</code> instead. |
| prob | Central credible-interval mass (default 0.95).                                                                                                             |

### Value

A data frame with ID, posterior `gebv` (mean), `sd`, `lower`, `upper`, ordered by decreasing `gebv`, with the pooled draws matrix (`nDraws`  $\times$  `nLevel`) attached as attribute `"draws"`.

**See Also**

`solution()`, the general replacement.

**Examples**

```
head(gebv(fit))
```

---

heritability

*Posterior distribution of heritability*


---

**Description**

Computes narrow-sense (genomic) heritability for every MCMC draw, returning the full posterior distribution rather than a single point estimate:

$$h^2 = \sigma_{genetic}^2 / (\sigma_{genetic}^2 + \sum \sigma_{other}^2 + \sigma_e^2).$$

**Usage**

```
heritability(fit, genetic = NULL, denominator = NULL, prob = 0.95)
```

**Arguments**

|                          |                                                                                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <code>fit</code>         | A <code>breedRB_fit</code> .                                                                                                  |
| <code>genetic</code>     | Character vector of genetic term identifiers (label or key) forming the numerator. Defaults to the <code>vm()</code> term(s). |
| <code>denominator</code> | Character vector of term identifiers summed in the denominator alongside <code>varE</code> . Defaults to all random terms.    |
| <code>prob</code>        | Central credible-interval mass (default 0.95).                                                                                |

**Value**

A list of class `breedRB_h2`: summary (mean/median/sd/CI) and draws (posterior vector of  $h^2$ ).

**Examples**

```
h2 <- heritability(fit); h2$summary; hist(h2$draws)
```

---

|                |                                                                      |
|----------------|----------------------------------------------------------------------|
| legendre_basis | <i>Orthogonal / orthonormal Legendre polynomial basis on [-1, 1]</i> |
|----------------|----------------------------------------------------------------------|

---

**Description**

Constructs a Legendre polynomial basis evaluated at a numeric vector scaled to  $[-1, 1]$ , up to a specified order, using the three-term recurrence. Optionally returns the orthonormal basis.

**Usage**

```
legendre_basis(x, order, orthonormal = FALSE)
```

**Arguments**

|             |                                                                             |
|-------------|-----------------------------------------------------------------------------|
| x           | Numeric vector (already scaled to $[-1, 1]$ ) where the basis is evaluated. |
| order       | Integer. Maximum polynomial degree. Returns columns for degrees 0..order.   |
| orthonormal | Logical. If TRUE, applies orthonormal scaling to each degree.               |

**Value**

A numeric matrix  $\text{length}(x) \times (\text{order} + 1)$ ; column  $j$  is the degree  $j - 1$  polynomial.

**Examples**

```
legendre_basis(seq(-1, 1, length.out = 5), order = 3)
```

---

|           |                                                                |
|-----------|----------------------------------------------------------------|
| mcmc_diag | <i>Markov-chain convergence diagnostics for a fitted model</i> |
|-----------|----------------------------------------------------------------|

---

**Description**

Summarises convergence of the variance-component chains: effective sample size, Geweke's  $z$ , and (when the fit has  $>1$  chain) the Gelman-Rubin potential-scale-reduction factor ( $\hat{R}$ ).

**Usage**

```
mcmc_diag(fit, what = "varcomp", plot = FALSE)
```

**Arguments**

|      |                                                                                                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fit  | A <code>breedRB_fit</code> .                                                                                                                                                                             |
| what | Passed to <code>as_mcmc()</code> (default "varcomp"; add "mu" to include the intercept).                                                                                                                 |
| plot | Logical. If TRUE, also render ggplot2 trace plots of the monitored chains (via <code>plot_trace()</code> ) and attach the plot object to the returned data frame as the "plot" attribute. Default FALSE. |

Value

A data frame with one row per monitored parameter: param, n\_eff, geweke\_z, and Rhat (NA for single-chain fits). When plot = TRUE, the corresponding ggplot is attached as attr(., "plot").

Examples

```
mcmc_diag(fit)
d <- mcmc_diag(fit, plot = TRUE)  # prints trace plots
attr(d, "plot")                  # the ggplot object
```

---

|             |                                                              |
|-------------|--------------------------------------------------------------|
| parse_model | <i>Parse asreml-style fixed / random / residual formulas</i> |
|-------------|--------------------------------------------------------------|

---

Description

Turns the three model formulas into an internal, engine-agnostic description used by [bbglr\(\)](#). Not usually called directly.

Usage

```
parse_model(fixed, random = NULL, residual = NULL)
```

Arguments

|          |                                                                                                                                                        |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| fixed    | Two-sided formula for the mean/fixed part, e.g. yield ~ 1 + env. Multi-trait models use cbind(t1, t2) ~ ....                                           |
| random   | One-sided formula for random terms, e.g. ~ vm(gen, G) + env:vm(gen, G). NULL for none.                                                                 |
| residual | One-sided residual formula. NULL (default) or ~ units gives homogeneous residuals; ~ dsum(~units   env) gives heterogeneous residual variances by env. |

Value

A breedRB\_terms list: response, fixed, random, residual.

Examples

```
parse_model(yield ~ 1 + env, ~ vm(gen, G))
```

---

|                |                                                                      |
|----------------|----------------------------------------------------------------------|
| plot_posterior | <i>Posterior density plot of variance components or heritability</i> |
|----------------|----------------------------------------------------------------------|

---

**Description**

Posterior density plot of variance components or heritability

**Usage**

```
plot_posterior(x, ...)

## S3 method for class 'breedRB_fit'
plot_posterior(x, ...)

## S3 method for class 'breedRB_h2'
plot_posterior(x, ...)
```

**Arguments**

|     |                                                             |
|-----|-------------------------------------------------------------|
| x   | A breedRB_fit (variance components) or a breedRB_h2 object. |
| ... | Unused.                                                     |

**Value**

A ggplot object.

---

|            |                                                     |
|------------|-----------------------------------------------------|
| plot_trace | <i>Trace plots of the variance-component chains</i> |
|------------|-----------------------------------------------------|

---

**Description**

Trace plots of the variance-component chains

**Usage**

```
plot_trace(fit, what = "varcomp")
```

**Arguments**

|      |                                       |
|------|---------------------------------------|
| fit  | A breedRB_fit.                        |
| what | Passed to <a href="#">as_mcmc()</a> . |

**Value**

A ggplot object (iteration on x, value on y, one panel per parameter, coloured by chain).



pr

*Posterior probability of ranking in the top fraction***Description**

For a model term, compute the posterior probability that each level ranks among the best threshold fraction of levels. Working from the pooled posterior draws (see [solution\(\)](#)), each MCMC draw is ranked independently and the top  $k = \text{round}(\text{threshold} \times n)$  levels are flagged; the reported probability is the fraction of draws in which a level is flagged. This is the Bayesian "probability of being in the top X" candidates by their whole posterior, not just their point estimate.

**Usage**

```
pr(fit, term = NULL, type = NULL, threshold = 0.2, higher = TRUE, pair = FALSE)
```

**Arguments**

|           |                                                                                                                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fit       | A <code>breedRB_fit</code> (single-trait).                                                                                                                                                                    |
| term      | Term identifier (label as written in the formula, e.g. "gen", or the internal key). If NULL, the first random term is used.                                                                                   |
| type      | Optional; one of "random" or "fixed", validated against the term's actual role. For a fixed term fitted with treatment contrasts the reference level is not among the coefficients and is therefore excluded. |
| threshold | Selected fraction in (0, 1); 0.20 = top 20%. Ignored when <code>pair = TRUE</code> .                                                                                                                          |
| higher    | Logical (default TRUE). If TRUE the "top" is the highest values (e.g. yield); set FALSE when smaller values are better. Ignored when <code>pair = TRUE</code> .                                               |
| pair      | Logical (default FALSE). If TRUE, return the pairwise table of posterior probabilities $P(A > B)$ for every pair of levels instead of the top-fraction summary.                                               |

**Details**

Because every draw is ranked on its own, adding the intercept `mu` would not change the ranking, so `pr()` is unaffected by centering.

**Value**

If `pair = FALSE`, a data frame with effect (level name), `solution` (posterior mean), and `prob` (posterior probability of ranking in the selected fraction), ordered by decreasing `prob`, with `k` and `threshold` attached as attributes. If `pair = TRUE`, a data frame with A, B, and `prob = P(A > B)` for every unordered pair, where A is the member with the larger posterior mean (so `prob >= 0.5` for well-separated pairs), ordered by decreasing `prob`.

**See Also**

[solution\(\)](#) for the underlying posterior draws.

## Examples

```
# probability each genotype is in the top 20%
pr(fit, term = "gen", type = "random", threshold = 0.20)
# pairwise probabilities P(A > B)
pr(fit, term = "gen", type = "random", pair = TRUE)
```

---

|          |                                                                     |
|----------|---------------------------------------------------------------------|
| solution | <i>Posterior solutions (BLUPs / fixed effects) for a model term</i> |
|----------|---------------------------------------------------------------------|

---

## Description

Extracts the posterior distribution of the effect solutions for any term in a fitted model, pooled across chains:

- **Random terms** return BLUPs. A `vm()` genomic term is back-mapped through its principal-component rotation to per-genotype values (identical to `gebv()`); a plain random factor (fitted without a relationship matrix) returns one solution per level; random regression / covariate terms return one solution per basis coefficient.
- **Fixed terms** return the posterior of each estimable coefficient (treatment contrasts; the reference level is the implicit baseline at 0).

## Usage

```
solution(fit, term = NULL, type = NULL, prob = 0.95, add_mu = FALSE)
```

## Arguments

|                     |                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>fit</code>    | A <code>breedRB_fit</code> (single-trait).                                                                                                                                                                                                                                                                                                                |
| <code>term</code>   | Term identifier (the label as written in the formula, e.g. "gen" or "env", or the internal key). If <code>NULL</code> , the first random term is used.                                                                                                                                                                                                    |
| <code>type</code>   | Optional; one of "random" or "fixed". When supplied it is validated against the term's actual role (a mismatch is an error), which is a convenient guard when a name could be ambiguous.                                                                                                                                                                  |
| <code>prob</code>   | Central credible-interval mass (default 0.95).                                                                                                                                                                                                                                                                                                            |
| <code>add_mu</code> | Logical (default <code>FALSE</code> ). If <code>TRUE</code> , the model intercept <code>mu</code> is added to every draw, shifting the solutions onto the overall-mean scale (e.g. <code>BLUP + mu</code> ). Most useful for random BLUPs; the addition is done per posterior draw so the credible intervals account for uncertainty in <code>mu</code> . |

## Value

A data frame with `effect` (level / coefficient name), `solution` (posterior mean), `sd`, `lower`, `upper`. Random-term rows are ordered by decreasing `solution`; fixed-term rows keep design order. The pooled draws matrix (`nDraws` x `nEffect`) is attached as attribute "draws", and the resolved role as attribute "type".

**See Also**

[gebv\(\)](#) for genomic breeding values from a `vm()` term.

**Examples**

```
solution(fit, term = "gen", type = "random")           # random BLUPs (no G matrix)
solution(fit, term = "gen", type = "random", add_mu = TRUE) # BLUP + mu
solution(fit, term = "env", type = "fixed")           # fixed-effect estimates
```

---

solve\_SNP

*Marker (SNP) effects from a genomic model*

---

**Description**

Recovers the posterior distribution of per-marker allele-substitution effects  $b$  on the centred-marker scale, such that the genomic breeding values satisfy  $GEV = M_c b$  (where  $M_c$  is the column-centred marker matrix). This lets a **GBLUP** fit — which estimates breeding values rather than marker effects — be back-transformed to marker effects:

$$b = M_c^\top (M_c M_c^\top)^{-1} u$$

applied to every posterior draw of the breeding values  $u$ . For a **RR-BLUP** fit (`mrk()` chose RR-BLUP because genotypes outnumbered markers) the marker effects are estimated directly and are simply rescaled and returned.

**Usage**

```
solve_SNP(fit, M = NULL, term = NULL, prob = 0.95)
```

**Arguments**

|                   |                                                                                                                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>fit</code>  | A <code>breedRB_fit</code> (single-trait) with a <code>vm()</code> or <code>mrk()</code> genomic term.                                                                                         |
| <code>M</code>    | Marker matrix with genotypes in rows (row names matching the genotype IDs) and markers in columns. Required for a GBLUP fit; optional for a RR-BLUP fit (marker names are taken from the fit). |
| <code>term</code> | Genomic term identifier (label or key). Defaults to the first genomic term.                                                                                                                    |
| <code>prob</code> | Central credible-interval mass (default 0.95).                                                                                                                                                 |

**Details**

The back-transform requires  $M_c M_c^\top$  to be invertible, which holds when the number of markers is at least the number of genotypes — exactly the regime in which `mrk()` selects GBLUP.

**Value**

A data frame with marker, effect (posterior mean), sd, lower, upper, in marker (design) order, with the pooled draws matrix (nDraws x nMarker) attached as attribute "draws" and the fitted method as attribute "method".

**See Also**

`gebv()`, `solution()`.

**Examples**

```
snp <- solve_SNP(fit, M) # back-solve marker effects from a GBLUP fit
head(snp[order(-abs(snp$effect)), ])
```

---

varcomp

---

*Posterior variance components of a fitted model*


---

**Description**

Returns the posterior distribution and summaries of every random-term variance and the residual variance, pooling all chains.

**Usage**

```
varcomp(fit, prob = 0.95, draws = FALSE)
```

**Arguments**

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| <code>fit</code>   | A <code>breedRB_fit</code> .                                    |
| <code>prob</code>  | Central credible-interval mass (default 0.95).                  |
| <code>draws</code> | Logical; if TRUE also return the pooled posterior draws matrix. |

**Value**

A data frame of per-component summaries (term, mean, median, sd, lower, upper), with the pooled draws attached as attribute "draws" when `draws = TRUE`.

**Examples**

```
varcomp(fit)
```